

# ECEN 662: Estimation & Detection Theory

## Lecture 1: Principle of Data Reduction and Sufficiency

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- In other words, does there exist  $T = T(X)$  of dimension  $m < n$  such that  $T$  carries *all* the useful information about  $\theta$ ?

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- In other words, does there exist  $T = T(X)$  of dimension  $m < n$  such that  $T$  carries *all* the useful information about  $\theta$ ?
- If so, for the purpose of studying  $\theta$ , we could discard the raw observation  $x$  and retain only the compressed statistic  $t = T(x)$

## Example: Bernoulli trials

- More specifically, let us consider the example of *Bernoulli trials*, in which  $X = [X_1, \dots, X_n]^t$ ,

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- Intuitively, to infer about  $\theta$ , it *suffices* to know how many 1's are in the realization of  $X$  - the positions of 1's appear to be rather irrelevant
- In another word, our intuition suggests that

$$T(X) = \sum_{i=1}^n X_i$$

is *sufficient* for inferring about  $\theta$ . But how do we justify this *probabilistically*?

## A formula for $p_{\theta}(x|t)$

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- We thus have

$$p_\theta(x) = p_\theta(x, T(x)) = p_\theta(T(x))p_\theta(x|T(x))$$

and hence

$$p_\theta(x|T(x)) = \frac{p_\theta(x)}{p_\theta(T(x))}$$

# Back to Bernoulli trials

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- It follows that

$$\begin{aligned} p_\theta(x) &= \prod_{i=1}^n p_\theta(x_i) \\ &= \prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i} \\ &= \theta^{T(x)} (1 - \theta)^{n-T(x)} \end{aligned}$$

where

$$T(x) = \sum_{i=1}^n x_i$$

- Further note that

$$T \sim \textit{Binomial}(n, \theta)$$

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- By the previous formula, for any  $(x, t)$  such that  $T(x) = t$  we have

$$p_{\theta}(x|t) = \frac{p_{\theta}(x)}{p_{\theta}(t)} = \frac{\theta^{T(x)}(1 - \theta)^{n-T(x)}}{\binom{n}{t} \theta^t (1 - \theta)^{n-t}} = \frac{1}{\binom{n}{t}}$$

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- Note that the above conditional distribution is *independent* of  $\theta$ , i.e, given  $T = T(X)$ , full knowledge of  $X$  brings *no* additional information about  $\theta$ . This is the *statistical* notion of *sufficiency* that we look for

# Sufficient statistic

- Formally, let  $X \sim f_\theta(x)$ . A statistic  $T = T(X)$  is said to be a *sufficient statistic* for  $\theta$  if the conditional distribution of  $X$  given  $T$  is *independent* of  $\theta$ , i.e., the functional form of  $f_\theta(x|t)$  does *not* involve  $\theta$

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- Recall that when  $X$  is discrete, we have

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$$p_\theta(x) = p_\theta(x, T(x)) = p_\theta(T(x))p_\theta(x|T(x))$$

- If the conditional distribution  $p_\theta(x|t)$  does *not* depend on  $\theta$ , the dependence of  $p_\theta(x)$  on  $\theta$  is manifested *entirely* in  $p_\theta(t)$ , and in this case, any inference strategy based on  $p_\theta(x)$  can be replaced by a strategy based on  $p_\theta(t)$

# Fisher-Neyman factorization theorem

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- (*Fisher-Neyman factorization theorem*) Let  $f_\theta(x)$  be a pdf or pmf of  $X$ . A statistic  $T = T(X)$  is sufficient for  $\theta$  *if and only if* there exist functions  $g_\theta(t)$  and  $h(x)$  such that:

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- In particular, if the likelihood  $f_\theta(x)$  depends  $x$  *only* through  $T(x)$ , then  $T = T(X)$  is a sufficient statistic for  $\theta$

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$$T(x) = \sum_{i=1}^n x_i,$$

so by the *Fisher-Neyman factorization theorem*,

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- Note that

$$p_{\theta}(x|t) = \frac{p_{\theta}(x)}{p_{\theta}(t)}$$

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- Since  $p_{\theta}(x|t)$  does *not* depend on  $\theta$ , by definition  $T = T(x)$  is a sufficient statistic for  $\theta$

## Another formula for $p_{\theta}(x|t)$

- If

$$p_{\theta}(x) = h(x)g_{\theta}(T(x))$$

the conditional distribution

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for any  $(x, t)$  such that  $T(x) = t$

- For Bernoulli trials, we have  $h(x) = 1$  for all  $x \in \{0, 1\}^n$  so

$$p_{\theta}(x|t) = \frac{1}{\sum_{y:\sum_{i=1}^n y_i=t} 1} = \frac{1}{\binom{n}{t}}$$

for any  $(x, t)$  such that  $\sum_{i=1}^n x_i = t$

## Example: Gaussian with unknown mean

- Suppose we observe  $X = [X_1, \dots, X_n]^t$ , where

$$X_i \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma^2)$$

$\sigma^2$  is a *known* constant, and  $\mu$  is the *unknown* parameter

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- We have

$$f_\mu(x_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

and hence

$$\begin{aligned} f_\mu(x) &= \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right) \\ &= \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(\frac{\mu}{\sigma^2} \sum_{i=1}^n x_i - \frac{n\mu^2}{2\sigma^2}\right) \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n x_i^2\right) \\ &= g_\mu(T(x))h(x) \end{aligned}$$

where

$$T(x) = \sum_{i=1}^n x_i$$

$$g_{\mu}(t) = \exp\left(\frac{\mu t}{\sigma^2} - \frac{n\mu^2}{2\sigma^2}\right)$$

$$h(x) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n x_i^2\right)$$

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- By the *Fisher-Neyman factorization theorem*,

$$T = \sum_{i=1}^n X_i$$

is a sufficient statistic for  $\mu$

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- By the *Fisher-Neyman factorization theorem*,

$$T(X) = \left[ \sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2 \right]^t$$

is a sufficient statistic for  $\theta = [\mu, \sigma^2]^t$

## Example: Uniform with unknown support

- Suppose we observe  $X = [X_1, \dots, X_n]^t$ , where

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- We have

$$f_{\theta}(x_i) = \frac{1}{\theta} 1_{\{x_i \leq \theta\}}$$

for any  $x_i \geq 0$  and hence

$$f_{\theta}(x) = \prod_{i=1}^n \frac{1}{\theta} 1_{\{x_i \leq \theta\}} = \frac{1}{\theta^n} \prod_{i=1}^n 1_{\{x_i \leq \theta\}} = \frac{1}{\theta^n} 1_{\{\max_{i \in [n]} x_i \leq \theta\}}$$

for any  $x \geq 0$ , where  $[n] := \{1, \dots, n\}$

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- Note that  $f_\theta(x)$  depends on  $x$  only through  $T(x) = \max_{i \in [n]} x_i$ . By the *Fisher-Neyman factorization theorem*,

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# Next lecture

Minimal and complete sufficient statistics